Activity 1

Create a model of a neuron using clay or beads.

15.1.2 Types of Neurons

The neurons may be of different types based on their structure and functions.

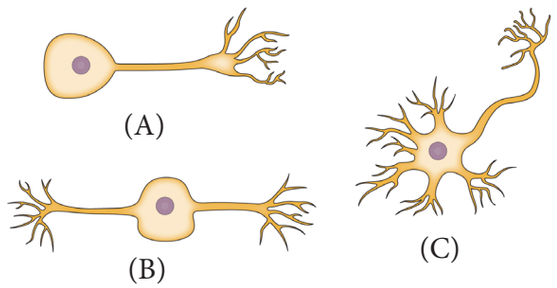


Fig. 15.2 Unipolar (A), Bipolar (B) and multipolar (C) neurons

Structurally the neurons may be of the following types:

(i) **Unipolar neurons**: Only one nerve process arises from the cyton which acts as both axon and dendron.

(ii) **Bipolar neurons**: The cyton gives rise to two nerve processes of which one acts as an axon while another as a dendron.

(iii) **Multipolar neurons**: The cyton gives rise to many dendrons and an axon

Unipolar Neurons	Found in early embryos but not in adult
Bipolar Neurons	Found in retina of eye and olfactory epithelium of nasal chambers
Multipolar Neurons	Found in cerebral cortex of brain

On the basis of functions neurons are categorised as:-

(i) **Sensory or afferent neurons** which carry impulses from the sense organ to the central nervous system.

(ii) **Motor or efferent neurons** which carry impulses from the central nervous system to effector organ such as the muscle fibre or the gland.

(iii) **Association neurons** conduct impulses between sensory and motor neurons.

15.1.3 Types of Nerve Fibres

Nerve fibres are of two types based on the presence or absence of myelin sheath.

(i) **Myelinated nerve fibre:** The axon is covered with myelin sheath

(ii) **Non-myelinated nerve fibre:** The axon is not covered by myelin sheath.

Myelinated and non-myelinated nerve fibres form the white matter and grey matter of the brain.

15.2 Transmission of Nerve Impulse

All the information from the environment are detected by the receptors located in our sense organs such as the eyes, the nose, the skin etc. Information from the receptors is transmitted as **electrical impulse** and is received by the dendritic tips of the neuron. This impulse travels from the dendrite to the cell body and then along the axon to its terminal end. On reaching the axonal end, it causes the nerve endings to release a chemical (**neurotransmitter**) which diffuses across a synapse and starts a similar electrical impulse in the dendrites of the next neuron, then to their cell body to be carried along the axon.

In this way, the electrical signal reaches the brain or spinal cord. The response from brain (or spinal cord) is similarly passed on to the effector organs such as the muscle or gland cell, that undergoes the desired response.

10th_Science Unit-15.indd 220 12/3/2022 3:02:58 PM